The Field Equations of the Screened Khronon Action, and a Verdict for Every Physics Class

Working note (PAPER70), draft v0.4, 2026-11-07. AI-assisted research programme; not peer reviewed. Not deposited. Companions: PAPER67 (DOI 10.5281/zenodo.23968171, the law and its failure record), PAPER68 (DOI 10.5281/zenodo.23990944, the action) and PAPER69 (DOI 10.5281/zenodo.24025170, the closure record). $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework.

Abstract

Gravity carries one extra scale, $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$, $\kappa = \frac{1}{2}$ fitted (proved not derivable), realised by the v3s screened khronon action. It passes Cassini, PPN, GW speed, galaxy lensing and SPARC, and fails AGC 114905, Malin 1 and the canonical FG1 tilt. It also under-fills X-COP clusters, groups, bright M31 dSphs and Coma UDGs by amounts that vary about 100× between classes, after the IMF accounts for the ellipticals and most of KiDS. Separately, it fails the CMB-lensing shape, which needs cold extra gravity at z 3400. No cold fluid, fixed-temperature sea, thermal light fermion or single measured key tried so far supplies either. The next lane must find one covariant quantity that sets that amplitude (not just a switch) in today's hot systems and is non-zero in the early universe, or show the two failures need separate causes. *(This is the checked seed paragraph of 2026-11-06, verbatim.)*

Since the seed, a joint systematics budget per class also explains the bright M31 dSphs, leaving X-COP, groups and the Coma UDGs as the measured boundary of the law; and a shock-entropy sector written into the action fails by the no-tuning rule, so every physics class on the record carries a label. This note writes down the action in the Legendre form every lane solved (correcting the form printed in PAPER68), its complete covariant field equations checked against direct variation, their 3+1 form and a convergent static solver. It then lists every gate with its frozen label, states the residue as measured numbers, and collects the three frozen tests that can kill the law. Four physical classes of extra ingredient were built and graded on both footings: a cold fluid, a warm fixed-temperature sea, a hot thermal fermion, and a shock-entropy sector written into the action. Each meets its own wall, and no candidate passes every gate. The most probable reading of the record is the deposited, scoped law plus separate per-class systematics, with the cluster, group and ultra-diffuse rows and the CMB wall left unexplained.

1 Conventions

Units $c = 1$ and $K = 4\pi G$ unless stated; restoring c: $K \to 4\pi G/c^2$, the prefactor of R is $c^4/16\pi G$, and $a_0 \to a_0/c^2$ inside W. Two footings are carried and never pooled: canonical $a_0 = 9.3603 \times 10^{-11}$ and alt $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. κ is fitted; a Lean proof shows that it is not derivable from the programme's premises. A gate is closed when it carries a label fixed by criteria committed before the first graded number; every lane runs a mutation twin that must fail.

2 The action, in Legendre form

PAPER68 prints $S = \frac{1}{4K}\int\sqrt{-g}\,[R - 2\mathcal{J}(Y) + 2\mathcal{K}(Q)] - \frac{\gamma_L}{2}\int\sqrt{-g}\,[\max(D\cdot P, 0)]^2 + S_m$. Read literally, P then appears only in the stiffness, its own equation forces $D\cdot P$ to be constant, and P decouples from gravity. The form every lane has actually solved puts $\mathcal{J}$ in Legendre form (QX17):

$$\frac{\mathcal{L}}{\sqrt{-g}} = \frac{R - 2\Lambda + 2\mu_K^2(Q-1)^2}{4K} + P^m A_m - W(K|P|) - \gamma_L F(\Pi) + \lambda\, n^m P_m, \qquad W(p) = \frac{a_0^2}{K}\Big[-\frac{p}{a_0} - \ln\Big(1 - \frac{p}{a_0}\Big)\Big], \tag{1}$$

with $F(\Pi) = \frac{1}{2}[\max(\Pi, 0)]^2$, $\Pi = D_m P^m$ and $\gamma_L = KL_\star^2/2$, plus minimally coupled matter S_m. Maximising over P returns the printed $\mathcal{J}$ exactly: $\max_P[P\cdot A - W(K|P|)] = (a_0^2/K)[u - \ln(1+u)]$ with $u = |A|/a_0$, attained at $K|P| = a_0 u/(1+u)$ (sympy, residual 0). So at $\gamma_L = 0$, Eq. (1) is the printed action; with $\gamma_L > 0$, P is an independent field and Eq. (1) is the theory. **This is an erratum on PAPER68's printed form only; no number changes.**

Definitions. τ is the khronon; $Q = \sqrt{-g^{mn}\partial_m\tau\,\partial_n\tau}$ ($= 1/N$ in unitary gauge); $n_m = -\partial_m\tau/Q$; $h_{mn} = g_{mn} + n_m n_n$; $l = \ln Q$; $A_m = n^n\nabla_n n_m = -h_m{}^n\partial_n l$; $D_m = h_m{}^n\partial_n$; $\Pi = \nabla_m P^m - P^m A_m$; $\Pi_+ = \max(\Pi, 0)$; $\hat{P} = P/|P|$; $\mathcal{D}_m = \partial_m\Pi_+ - \Pi_+\partial_m l$.

3 The field equations (QX17, CERTIFIED)

$$\text{(E1)}\quad n^m P_m = 0, \tag{2}$$

$$\text{(E2)}\quad (1+\gamma_L\Pi_+)A_m + \gamma_L D_m\Pi_+ = KW'(K|P|)\,\hat{P}_m, \qquad \lambda = (1+\gamma_L\Pi_+)\,n\cdot\partial \ln N + \gamma_L\, n\cdot\partial\Pi_+, \tag{3}$$

$$\text{(E3)}\quad \nabla_m\big[n^m\rho_Q - \lambda P^m/Q\big] = 0, \qquad \rho_Q = \frac{\nabla_m[(1+\gamma_L\Pi_+)P^m]}{Q} + \frac{\mu_K^2(Q-1)}{K}, \tag{4}$$

$$\text{(E4)}\quad G_{mn} + \Lambda g_{mn} = 2K\big(T^{\rm mat}_{mn} + T^{Q}_{mn} + T^{W}_{mn} + T^{P}_{mn}\big), \tag{5}$$

with

$$T^{Q}_{mn} = g_{mn}\frac{\mu_K^2(Q-1)^2}{2K} + \frac{\mu_K^2}{K}Q(Q-1)\,n_m n_n, \qquad T^{W}_{mn} = -g_{mn}W + KW'\,\frac{P_mP_n}{|P|},$$

$$T^{P}_{mn} = g_{mn}(P\cdot A) + 2P_{(m}\partial_{n)}l + (\nabla\cdot P)\,n_m n_n - 2\lambda\, n_{(m}P_{n)} + \gamma_L\big[-g_{mn}F + 2Fh_{mn} + h_{mn}P^r\mathcal{D}_r - 2P_{(m}\mathcal{D}_{n)}\big].$$

(E5) Matter is minimally coupled, $\nabla^m T^{\rm mat}_{mn} = 0$. The energy density seen by the foliation is $n^m n^n T_{mn} = (\mu_K^2/2K)(Q^2-1) + W + \Pi + \gamma_L F$. At $\gamma_L = 0$ the leaf part of (E2) inverts to $K|P| = a_0|E|/(a_0+|E|)$, the Legendre map; Π_+ is C^1, so the equations hold in the weak sense across the switch surface.

Checks. Each equation is compared with a direct Euler–Lagrange variation of Eq. (1) on jet symbols (mpmath, 40 digits, three random configurations in 2+1 dimensions): P equation max rel 3.8×10^{-41}, khronon equation 1.8×10^{-41}, full stress tensor 1.2×10^{-40}; FRW reduction exact. The MUTATE twin (two flipped signs) fails at 0.17–0.78.

AC2q (candidate, not adopted; QX17A, CERTIFIED). The dark-energy scalar is the clock, there is no $\mathcal{K}(Q)$ term, and the switch acts on the khronon charge $\Sigma = \Pi + Q^2$: $\mathcal{L}/\sqrt{-g} = R/4K + X - V(\phi) + P\cdot A - W - \gamma\{F(\Pi+2X) - (2X)^2/2\} + \lambda n\cdot P$, $X = Q^2/2$. Its P equation is (E2) with $\Pi_+ \to \Sigma_+$; its clock equation is $\nabla_m[n^m\rho_\phi - \lambda P^m/Q] = -V'(\phi)$ with $\rho_\phi = J_P/Q + Q[1-2\gamma(\Sigma_+ - Q^2)]$, which is $\Box\phi = V'$ at $P=\gamma=0$; at $P=0$ the stiffness stress vanishes identically, so the background is the canonical scalar's. The MUTATE twin fails at $0.23/0.39/7.3\times10^{-3}$.

4 3+1 form, reductions and solutions

ADM (QX17B). P and N carry no time derivatives, so the P equation and the lapse equation are constraints on each leaf. The P sector's momentum density vanishes on shell, so the momentum constraint is GR's; the tensor principal part is GR's, so $c_T = c$. The Lean kernel file for these steps has 12 theorems, standard axioms only, and its MUTATE twin fails. The lane as a whole is not certified: its numerical ADM row failed as frozen on a sign convention in the checker's off-shell momentum density, and its `solve_bvp` spherical solver did not converge.

Reductions. FRW with $P=0$: $\bar{Q} = 1 + I_0/(2\mu_K^2 a^3)$ and a dust-like $\rho \propto a^{-3}(1+w_0/a^3)$. Static, weak, $\gamma_L = 0$: $\nabla\cdot[\mu(|\nabla\Phi|/a_0)\nabla\Phi] + \mu_K^2\Phi = 4\pi G\rho_b$, $\phi = \psi$. Spherical, $\mu_K = 0$: the law $g_N = g^2/(g+a_0)$ (Lean). Static, $\gamma_L > 0$: the Cassini hedgehog inside $L_\star$. AC2q FRW: the lapse–polarization mixing is $(1-z)$ with $z = \gamma\dot{\phi}^2/\bar{N}^2$, so eliminating P multiplies the induced lapse term by $(1-z)^2$, not the legacy $(1-2z)^2$ (which is the exact factor of a different action). The UV speed is $c^2 = z/(1-z+z^2)$, and $0 < c_g \le c_{\rm ph} \le 1$ holds for any $f \ge 0$, so FRW health stands (FRW reconciliation chain: sympy ADM reduction and Lean, 11 theorems, standard axioms only; MUTATE 3/3 fail). The P block couples to the lapse only. The static-limit offset is half the earlier value, bounded by 1.1×10^{-10}. Still open: the AC7 Q2 / AC10–AC15 post-Newtonian chain carries the legacy factor and has not been rederived.

Solutions (QX17C, CERTIFIED as a solver). A conservative finite-difference Newton solver with a decaying Bessel outer boundary converges in 1–6 steps on both footings, about 5 s in all. It reproduces the law at $L_\star \to 0$ (worst 3.9×10^{-7}), the solar hedgehog to 7.1×10^{-6} of $g_\star = GM/L_\star^2$, and a $10^{10}M_\odot$ galaxy on the law to 4.8×10^{-6}; full-GR minus weak-field is 4.1×10^{-6} (Sun) and 5.4×10^{-7} (galaxy). An isolated point Sun reaches only 0.71/0.67 of the law at $10^3 L_\star$, so whether a galaxy built from screened stars shows the law is open.

5 Every gate, with its frozen label

The full table, one row per gate, is closure_2026_11_06/CLOSURE_FINAL.md; PAPER69's verdict table is its deposited form and is carried unchanged. The rows below are the 2026-11-06 lanes that came after PAPER69.

Gate	Label	Lane
Equations and health		
Covariant field equations, v3s and AC2q	CERTIFIED vs direct variation	QX17, QX17A
3+1 form, Lean kernels	Lean PASS; lane NOT CERTIFIED (numerical row, solver)	QX17B
Static spherical solver	CERTIFIED, both footings	QX17C
Π^6 repair, curved 3D spherical switch	PASS, not adopted; non-radial modes untested	BL3
AC2q decelerating slow roots	HARMLESS, scoped; $\epsilon_\phi = 2$ NOT REACHED by any record history	EP1, EP2
Nonlinear fold, shock-capturing	NO VERDICT as frozen (viscosity check fails); no pair where AC2q regularizes	FS1
Nonlinear fold, viscosity-free reading	DIFFERS, both footings: the 80 clock-only pairs fold 1.2 % earlier (DIFF-TIME EARLIER, edge-carried); the soft and phantom pairs (160) read SAME-REACH; no row is REGULAR and no pair shows AC2q regularizing; all checks pass	FS2
Moving Sun convergence	NO VERDICT, bracketed; largest time-mean on any rung 0.147 of the line. MS3 is running; its reading becomes the FINAL moving-Sun label (the last iteration)	MS1, MS2, MS3
Light-cone causality	FAIL (physical, far below data); no repair in the khronometric class	AC18, SV2
All-gates battery (static gates G1–G8; G9 CMB)		
Six candidates on G1–G8	NOTHING PASSES ALL STATIC GATES; every candidate fails the Coma UDGs by at least $3.97\,\sigma_U$	GB1
Cold fluid: $\mathcal{K}(Q)$ dust, amplitude freed	FAIL: passes the background and CMB lensing, fails G1–G7; cosmic capture 4.111 in every class	KD1
Cold component decaying late	FAIL: CMB-lensing χ^2 46.3/53.0 against 18.48 at the longest lifetime discs allow	LD1
Hot thermal fermion, 11.1377 eV	FAILS 7 of 9 gates; CMB-lensing χ^2 166.0–194.2	HC1
Shock-entropy field (proxy)	FAIL: X-COP and groups; need falls outward while the measured adiabat rises	EN1
Shock-entropy sector in the action	FAIL by the no-tuning rule; static solutions isentropic (f $\leq$ 0.04)	SE1
The step as several causes		
Multi-cause step	MULTI-CAUSE IN PART: IMF explains the ellipticals and KiDS	ST1
Joint systematics budget per class	IN PART: ellipticals, KiDS and bright M31 dSphs EXPLAINED; X-COP +0.440, groups +0.232, Coma UDGs +1.120 NOT EXPLAINED (canonical medians); CMB SEPARATE	SE2
Environment, re-read	NO: no environment key passes; satellites need about 1.6× the phantom, Coma UDGs 4–46×	EV1
Disc-only exclusion	every direction DEAD; even a perfect (oracle) exclusion fails GB1	DX1
One action-native scalar for both walls (OQ2 kept by the owner)	NO VERDICT as frozen. At the frozen fallback $N_{\min} = 8$ the S1 reading is DEAD on both footings, every start: SPARC vs groups at $\log q$ 3.0–3.4, gap up to 0.497 / 0.483 dex. Web row YES: the CMB wall needs the law's phantom off on the web ($F_{\mathrm{web,max}}$ 0.00481 / 0.00435). SEPARATED NO, as frozen	OQ1, OQ2
Outer edge of the law from the switch	NO EDGE IN THE ACTION: for an isolated galaxy the static switch variable is $\Sigma/X = 2 + \rho_{\mathrm{ph}}/\rho_{\mathrm{DE}} \geq 2$, so the phantom halo runs on until an external field ends it (the standard EFE radius, already graded)	outer-edge check
Owner-opened test lanes, 2026-11-07 (adopt nothing)		

Gate	Label	Lane
Host-held dust with the measured EFE capping the wells	FAIL (clusters, 0/20 cells, both footings); with no dust the EFE moves the KiDS baseline misfit to +0.870	QX7 iter_6
Remnant tracing the hot gas	SHAPE FAIL, both footings: the missing mass is more concentrated than the hot gas	QX7 iter_7
Asphericity of hydrostatic masses in the law's field	NEGLIGIBLE (≤ 0.005 dex, wrong sign), both footings	QX32

6 What an all-gates ingredient would have to supply

On top of the law, the extra gravitating mass as a fraction f of the baryons (KD1 diagnostic, canonical / alt):

Where	Needed f
SPARC discs	≤ 0.33 / 0.19
X-COP clusters	~ 4.2 / 3.8
groups	1.2–3.5 / 0.9–2.9
X-ray ellipticals	~ 0.9 / 0.8 (about half of it IMF)
dSph satellites	~ 1.7 / 1.3
Coma UDGs	~ 80
the CMB	cold at z 3400 ($w \leq 0.01$), lensing $\chi^2 \leq 18.5$ at $L \leq 400$

7 Four walls

1. **Cold fluid** (KD1, QX3, QX7). Every well captures it, discs included. The CMB and BBN need $\mu_K \geq 8.9 \times 10^6\,\mathrm{Mpc}^{-1}$; galaxies allow at most 79.
2. **Warm fixed-temperature sea** (QX12). Retention $\exp(\mathrm{depth}/c_s^2)$ is too sharp; the temperature would have to track each host.
3. **Hot thermal fermion** (HC1, HF1, VW1). Hot enough to stay out of discs, it under-fills clusters, cannot fill dSphs or UDGs, and fails CMB lensing; cold enough for the CMB, discs capture it. No window.
4. **Shock-entropy sector** (SE1). A conserved, non-cooling sector heated by its own shocks either sits isentropic, and holds almost nothing, or needs a different entropy history for each class at the same entropy. Its conserved rest mass also caps a host near the cosmic share, against the ~ 80 the Coma UDGs need.

The common reason: an ingredient that tracks each host's virial state has to virialise with the host; if it is also cold early, nothing keeps it out of discs. And the CMB needs the extra gravity at $z \simeq 3400$, where nothing is virialised, so any key that is on only in today's hot systems is off exactly where the CMB needs it on.

8 The residue: the law's measured boundary

SE2 draws every systematic knob jointly per class (IMF, velocity-dispersion scale, distance, anisotropy, membership, 3-D position, tidal admissibility). The posterior step of the plain law, in dex (median, 95 % interval):

Class	canonical	alt	label
X-COP clusters	+0.440 [+0.372, +0.484]	+0.408 [+0.341, +0.451]	NOT EXPLAINED
groups	+0.232 [+0.078, +0.362]	+0.194 [+0.041, +0.324]	NOT EXPLAINED
Coma UDGs (9)	+1.120 [+0.541, +1.718]	+1.075 [+0.472, +1.673]	NOT EXPLAINED
X-ray ellipticals	+0.067	+0.050	EXPLAINED
KiDS early/late	+0.085	+0.084	EXPLAINED
bright M31 dSphs (21)	+0.228	+0.219	EXPLAINED
CMB-lensing shape	cold extra gravity at z 3400; no key passes the hot classes		SEPARATE

The groups row is the weakest (z 3.04 / 2.55). Gas in the Coma UDGs is not in the budget and would lower their step; hydrostatic bias is set to 0 and would raise the cluster and group steps. These numbers are the boundary of the law as measured, not a mechanism.

9 Kill tests, frozen

Three tests can kill the law with data not yet in hand. Each is frozen before the data; nothing else is pending.

1. **Gaia DR4 wide binaries: Newton.** v3s screens binaries, so it predicts Newtonian motion: the frozen pipeline returns γ_∞ 1.0019 / 1.0100 (canonical / alt), not separable from Newton and separable from plain QUMOND (1.1494 / 1.2081). The prediction is sealed (CQ2, SHA-256 1d65509dbf). Scored as the screened (medium-first) reading of the frozen DR4 rows (owner decision 413c1c8e24), v3s is excluded iff $\hat{\gamma} - 1 \geq 3\sigma$, with the gate $\sigma_{\rm sys} < 0.048$ / 0.058 and the R rule (KILL iff $R - 1 \geq \max(0.01, 3\sigma_R)$). At the registered error (N 30,000, $\sigma_{\rm sys}$ 0.020) v3s is EXCLUDED for $\hat{\gamma}$ in 1.083–1.23; at the realistic N of about 8,000 the edge moves up.
2. $a_0(z)$ **flat to** z **2.33.** a_0 tracks $\rho_{\rm DE}$ (CFG689 quantiles, BAO + SN, no ΛCDM input). No sample on disk can decide it: all 52 cells are BLOCKED, because each sample's common mode (0.30–0.60 dex) is 4–11× the largest $e_{3\sigma}$ (0.084 dex at z^* 2.33). It needs a proposal-class ALMA sample with a same-estimator $z < 0.1$ reference (AT2). It kills the law if, at a powered z^*, the rival is favoured or neither model fits (AT1's rule).
3. **Lensing = dynamics.** v3s predicts $M_{\rm lens}/M_{\rm dyn} = 1$ to $\leq 6.5 \times 10^{-5}$ at 310 kpc. KiDS isolated lenses give $A = 1.129 \pm 0.206$ (canonical) and 1.035 ± 0.187 (alt), PASS but with weak power (V3W W3) The kill line is the frozen grade: FAIL if the KiDS ratio sits more than $2\sigma_{\rm tot}$ from 1, or the matched-host ratio more than $3\sigma_{\rm tot}$.

The exact statistics and kill lines are collected in closure_2026_11_07/PREDICTIONS_2026_11_07.md (52d4f27840), checked by its own script.

10 Reading

Every physics class on the record now carries a label, and none passes every gate. The law, and the action within its scope, describe cold rotating discs and the Solar System. The ellipticals, KiDS and the bright M31 dwarfs are accounted for by measured systematics. X-COP, the groups and the Coma UDGs are not, and the CMB-lensing wall stands on its own. This is the best remaining answer the record supports, not a closed theory.

11 What waits on the owner

The PAPER68 erratum for the Legendre form; adoption of the Π^6 stiffness repair; the deposit of this note. Still running or not yet landed: MS3 (its reading becomes the FINAL moving-Sun label) and the owner-opened test lanes WL1 and EW1–EW3. Their labels will be appended when they land. They adopt nothing, and none changes a row above unless its frozen criteria say so.

Method

Every lane froze its criteria in a commit before any script ran, and every lane has a MUTATE twin that must fail. Footings are reported apart. No ΛCDM inputs were used. Scripts and outputs are in the repository under closure_2026_11_06/ and closure_2026_11_07/ at the commits the lane RESULT files name.